

Understanding the Strategy Logic

Moving average crossover strategies are the "Hello, World!" of algorithmic trading—a perfect starting point for beginners. The concept is refreshingly straightforward: we use two moving averages of different lengths to identify potential trend changes.

Think of it like watching two cars traveling along the same road at different speeds. The faster car (short-term moving average) is more responsive to the changing landscape, while the slower car (long-term moving average) maintains a more steady course. When the faster car overtakes the slower one, it suggests the road is now trending upward. When the faster car falls behind the slower one, it indicates the road might be trending downward.

Here's how the strategy works in trading terms:

1.
Calculate two moving averages of the price—one short-term (e.g., 20 days) and one long-term (e.g., 50 days)
2.
Buy Signal: When the short-term moving average crosses above the long-term moving average
3.
Sell Signal: When the short-term moving average crosses below the long-term moving average
- 4.

This strategy aims to capture trends while filtering out short-term market noise. It's based on the idea that price movements often exhibit momentum—they tend to continue in the same direction for some time before reversing.

Let's see what this looks like in Python code:

```
1. import pandas as pd
2. import numpy as np
3. import yfinance as yf
4.
5. # Download historical data for Apple
```